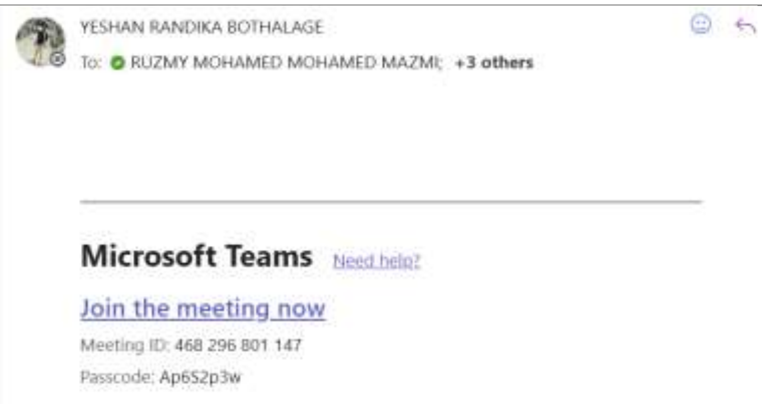
Individual Work Log

PROJECT NAME:	AI tool with Mistral 24B to extract knowledge Graph components from Research Papers
STUDENT NAME:	**Chiraath Madahapola**
STUDENT ID:	**104834009**

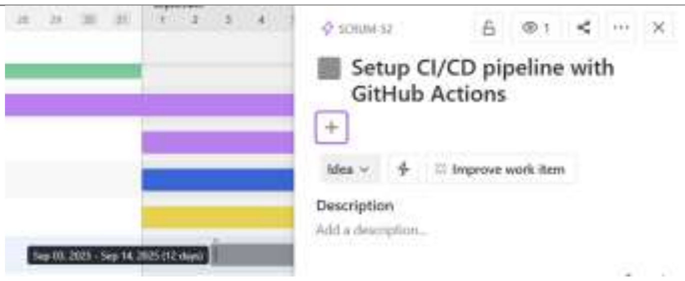
WEEK #	**5**		
Dates covered:	01/09/2025 - 07/09/2025		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Meeting with supervisor	Completed	**45 min**	
Teem meetings + individual meetings with group leader (in person and online)	Completed	7h	

<table>
<tr><td>worked on the
PDF text
extraction
module with
Ruzmy.</td><td>completed</td><td>5
hours</td><td>
# Embedding generation + FAISS index
embeddings = OpenAIEmbeddings(model=EMBED_MODEL, api_key=OPENAI_API_KEY)

faiss_store = FAISS.from_documents(chunks, embeddings)

INDEX_DIR = "/content/faiss_index"
faiss_store.save_local(INDEX_DIR)

print(f"FAISS index saved to {INDEX_DIR}")

</td></tr>
<tr><td>Setup CI/CD
pipeline with
GitHub Actions</td><td>Pending</td><td>4
hours</td><td>
</td></tr>
<tr><td>Reading about
Semantic
Search
Implementation
with FAISS</td><td>Reading</td><td>5
hours</td><td>You’ve learned how to use a FAISS semantic search pipeline to retrieve the most relevant text chunks from your document database. By writing this code:

• You created a query to ask in natural language.
• You used <code>faiss_store.similarity_search(query, k=8)</code> to get the top 8 most similar results based on vector embeddings (not just keyword matching).
• You accessed each document’s metadata (source, page, chunk ID) and content to see where the information came from.
• You learned how to loop through and print results in a clean format.

In short, this code shows you how to search documents semantically, view where results came from, and understand how vector stores like FAISS organize and retrieve information.</td></tr>
</table>

TOTAL WEEKLY TIME SPENT		**21.45**	

Summary/weekly reflection for Week 5:

Key Tasks Completed:

- Understood the benefits of containerization with Docker for consistent and reproducible deployment environments.
- Conducted individual team member meetings to monitor progress on Mistral 24B research tasks and delegate new assignments.
- Sent a follow-up email to the client to clarify requirements for Mistral 24B integration and address delayed responses.
- Reading about Semantic Search Implementation with FAISS

Pending Task:

Setup CI/CD pipeline with GitHub Actions to automate testing and deployment for

Mistral 24B API integration.

Technical Learning & Development:

- Deepened understanding of Neo4j integration with the backend system to support Mistral 24B data handling.
- Explored deployment options for the Mistral 24B model, focusing on scalability and performance.
- Researched GitHub Actions workflows to prepare for CI/CD pipeline implementation for Mistral 24B.

Literature Review:

- Conducted an in-depth analysis of Mistral AI documentation: *Mistral AI Models Overview*.

- Focused on Mistral 24B's advanced features, optimization techniques, and API integration strategies.
- Reviewed best practices for CI/CD pipelines to ensure compatibility with Mistral 24B deployment requirements.

Challenges & Considerations:

- **CI/CD Implementation:** Limited progress on setting up the CI/CD pipeline with GitHub Actions due to prioritization of API exploration; planning to allocate dedicated time next week.

Next Steps:

- Establish regular communication channels for deployment updates and begin monitoring CI/CD pipeline performance.

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Model Selection for Project	12/09/2025
Testing the selected model	14/09/2025